

Does Language Model Behavior Reproduce Racial Bias in Clinical Pain Management? An Audit of AI Value Alignment in Healthcare Recommendations

Team: Smb (Year, Major) · [Groupmate 2] (Year, Major) · [Groupmate 3] (Year, Major)

Team Responsibilities: Smb – vignette design, LLM API integration, running the audit and quantitative scoring; [Groupmate 2] – literature review, theoretical framework, gap analysis; [Groupmate 3] – results visualization (table/chart), ethical implications and execution-risk analysis.

1. Research Thesis and Course Connections

Central Question

Does an existing large language model recommend different pain-management approaches for clinically identical patient vignettes when only the patient's stated race varies? This matters because LLMs are increasingly proposed for clinical decision support, documentation, and patient-facing communication; if a model reproduces the well-documented human bias of under-treating pain in Black patients (Hoffman et al., 2016), it risks automating and scaling an existing healthcare inequity rather than correcting it.

Field Landscape

This project connects to course concepts including (1) *value alignment as behavioral consistency* – whether a model's outputs align with the equitable-care values it is presumed to embody regardless of demographic framing; (2) *bias measurement via controlled prompt audits*, the methodological approach used in course discussions of benchmark-based fairness evaluation; and (3) *disparate impact without disparate intent* – a model need not be designed to discriminate to produce discriminatory outputs, a recurring theme in our course's treatment of algorithmic harm.

Academic Contribution

Beyond building an audit pipeline, this project contributes a direct experimental bridge between the classic human-clinician bias literature (vignette-based studies like Hoffman et al., 2016) and LLM-bias literature, using the same vignette-design logic to test whether an AI system trained on human-generated text reproduces a specific, well-established human bias rather than bias in general.

2. Literature and Theoretical Framework

Project-specific sources: Hoffman, K.M., Trawalter, S., Axt, J.R., & Oliver, M.N. (2016). Racial bias in pain assessment and treatment recommendations. *PNAS*, 113(16), 4296–4301. **External sources (beyond class readings):** Omiye et al. (2023, *npj Digital Medicine*), "Large language models propagate race-based medicine"; Zack et al. (2024, *Lancet Digital Health*), "Coding Inequity: Assessing GPT-4's Potential for Perpetuating Racial and Gender Biases in Healthcare"; Parrish et al. (2022), the BBQ bias-benchmark paper, as a methodological reference for constructed bias-probing datasets.

Theoretical approach: We combine an algorithmic-fairness lens (disparate impact, measured via outcome differences across protected-attribute-varied inputs) with alignment theory's framing of "faithful representation of intended values" – here, equitable care – as a target the model may fail to meet even without explicit discriminatory design.

Gap analysis: Zack et al. (2024) tested GPT-4 broadly across diagnosis, imaging, and subjective assessment tasks, and found race-linked differences in opioid-misuse suspicion; Omiye et al. (2023) tested explicit race-based medical misconception questions. Neither directly replicates Hoffman et al.'s (2016) specific pain-severity vignette design, and both focus on a single proprietary model. Our project fills this gap by applying Hoffman et al.'s own vignette structure directly to an LLM, providing a tighter human-to-AI comparison than prior LLM-bias audits.

Integration: Hoffman et al. supplies our vignette methodology and outcome variable (pain-treatment recommendation strength); Omiye et al. and Zack et al. establish that LLM race-linked bias in clinical contexts is real and measurable, motivating our narrower replication; BBQ supplies the constructed-dataset audit paradigm our Mini-Audit follows.

3. Methodology and Research Design

Model: An existing, already-trained LLM accessed via free-tier API (e.g., a current commercial or open-source chat model) – no training or fine-tuning. **Dataset (hand-labeled, <50 items):** ~15–20 base clinical pain vignettes (varying pain type/severity, adapted from Hoffman et al.'s vignette structure), each rendered in 3 race-coded versions (Black / White / Hispanic patient) with all clinical details held identical, yielding ~45–60 total prompts. **Metric:** pain-treatment recommendation strength (ordinal: no medication suggested / over-the-counter / prescription non-opioid / opioid-class), and refusal/hedge rate. **Comparison groups:** 3 race-coded patient conditions. **Sample size:** ~45–60 prompts, each run once through the model. **Output:** one summary table (mean recommendation strength and hedge rate per group), one bar chart visualizing the gap, and a paragraph interpreting whether observed differences replicate the human-bias pattern found in Hoffman et al. (2016).

4. Ethical and Societal Implications

If disparities are found, this signals risk for any downstream deployment of LLMs in clinical decision support, patient triage assistance, or even informal patient use, since bias here would affect real pain-treatment decisions at scale. Stakeholders include patients (unequal care), clinicians (over-reliance on biased AI suggestions), and deploying institutions (liability, equity obligations). Post-deployment, we recommend mandatory subgroup bias audits before clinical integration, transparent reporting of known disparities to clinician end-users, and human-in-the-loop review for pain-related recommendations specifically, given the documented severity of this bias in the human baseline literature.

5. Open Questions and Execution Risks

Open questions: whether a small (~50-prompt) sample can detect a reliable effect given LLM response variability (Omiye et al. found inconsistent repeated responses to identical prompts); whether observed differences (if any) reflect genuine bias versus prompt-wording artifacts. Execution risks: API access/rate limits within our two-week window; potential model refusal on medication-recommendation prompts entirely, which we will treat as a codeable outcome (refusal rate) rather than a failure; and the need for inter-rater agreement if recommendation strength is manually coded from free-text responses.

References

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- Zack, T., Lehman, E., Suzgun, M., Rodriguez, J. A., Celi, L. A., Gichoya, J., Jurafsky, D., Szolovits, P., Bates, D. W., Abdulnour, R. E., Butte, A. J., & Alsentzer, E. (2024). Coding inequity: assessing GPT-4's potential for perpetuating racial and gender biases in healthcare. *Lancet Digital Health*, 6(1), e12–e22.
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